

- 8 **C** In Fig. 9., the vector sum of the tension force and the weight provides **the centripetal force** to allow the mass to move in circular motion.

$$T - mg = ma_c$$

$$T = (mg + ma_c) > mg$$

The tension force pulling the mass upwards is equal and opposite to the tension force pulling the balance downward. (Both form the tension in the string)

The balance reading is given by $T / g > m$ i.e. $> 50 \text{ g}$